LECTURE 8

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$S_n = a_1 + a_2 + \dots + a_n$

$\{S_n\}$ $\quad S_1 \le S_2 < S_3 \dots \uparrow$ ing

$S_n \uparrow$ ing $\begin{cases} \text{bdd above} \to \text{cgt MCT} \\ \text{unbdd above} \to \text{div} \end{cases}$

$\{S_n\}$ cgt $\quad \sum a_n$ cgt

div $\quad \sum a_n$ div.

Eg. $\sum n^2 = \sum a_n$

$$S_n = \frac{n(n+1)(2n+1)}{6}$$

Eg. $\sum r^n$

$$S_n = r + r^2 + \dots + r^n$$

$$= \frac{r(1-r^n)}{1-r}$$

$\underline{\underline{|r| < 1}}$

$$\lim_{n\to\infty} S_n = \frac{r}{1-r}$$

$\left.\frac{1}{2^n} \to \infty\right\} \to 0$

$r = 1 \quad S_n = \underbrace{1 + 1 + 1 + \dots + 1}_{n \text{ terms}} = n$

$\lim S_n \to \infty$

$r = 1 \quad \sum r_n \rightarrow$ div

$|r| < 1 \quad \sum r_n \rightarrow$ conv.

Eg. ① $\sum \frac{1}{5n^2+7n+11}$ ② $\sum \frac{1}{2n^2-n}$ ③ $\sum \frac{1}{n+\sqrt{n}}$

convergence test:

Test 1: p-series test.

$\sum \frac{1}{n^p}$ conv. $p > 1$

div $p \leq 1$

Eg. $\sum \frac{1}{n} \rightarrow$ div

Imp. Remark: for $\sum a_n$ to converge

$\lim a_n = 0$

Eg. 1) $\sum \frac{1}{n^{3/2}}$

2) $\sum n \sin \frac{1}{n}$

$a_n = \frac{\sin 1/n}{1/n}$

$\lim a_n = 1$